

Autonomy Dimension of Graphs: Spectral Separation, Product Activation, and Entropy Bounds

October 7, 2026

Abstract

An autonomous binary coordinate on a graph is a two-coloring for which the probability that a simple random walk changes color depends only on its present color. We study the least number of such coordinates needed to distinguish all vertices, called the autonomy dimension, with value infinity when no separating family exists. We give a spectral criterion for finiteness and prove two structural results. First, $\text{adim}(C_5) = \infty$ but $\text{adim}(C_5 \square C_5) = 6$: taking a Cartesian product can create a separating family even when neither factor has one. Second, if W_r consists of r copies of K_4 sharing one vertex, then

$$\text{adim}(W_r) = \frac{\log_2(3r)}{H_2(1/3)} + O(\log \log r).$$

Thus exact autonomous observation has an asymptotic cost strictly above the information bound for naming vertices. We also classify cycles, establish a complement invariance for regular graphs, and show that finite autonomy dimension is generically destroyed by arbitrary positive edge weights. The elementary proofs are accompanied by exact finite checks. Individual coordinates are classical binary strong lumpings; the proposed organizing invariant is the minimum size of a separating family.

Keywords. Strong lumpability; perfect two-coloring; graph spectrum; Cartesian product; separating family; binary entropy.

1 Definition and purpose

Throughout, $G = (V, E)$ is a finite connected simple undirected graph with $n \geq 2$ vertices, unless weights are explicitly introduced. Let P be its simple random walk operator,

$$(Pf)(v) = \frac{1}{\deg(v)} \sum_{w \sim v} f(w), \quad \pi(v) = \frac{\deg(v)}{2|E|}.$$

The question is whether a vertex can be described completely by a small number of binary observations, each of which has closed Markov dynamics.

Definition 1.1. A nonconstant function $f : V \rightarrow \{0, 1\}$ is *autonomous* if there are constants $a, b \in [0, 1]$ such that

$$\frac{|\{w \sim v : f(w) \neq f(v)\}|}{\deg(v)} = \begin{cases} a, & f(v) = 0, \\ b, & f(v) = 1. \end{cases}$$

The *autonomy dimension* $\text{adim}(G)$ is the least d for which there exist autonomous $f_1, \dots, f_d$ making $v \mapsto (f_1(v), \dots, f_d(v))$ injective. If no such family exists, set $\text{adim}(G) = \infty$.

For a simple random walk (X_t) , an autonomous f makes $(f(X_t))$ a Markov chain, for every initial distribution, with the same transition matrix

$$\begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

Indeed, conditioning further on the current vertex does not change the prescribed transition probabilities. Conversely, requiring this same matrix for walks started at every vertex gives Definition 1.1 at the first step. Connectedness implies $a, b > 0$.

Each coordinate therefore predicts its own next-step law from its current value alone. Different coordinates need not be statistically independent, and their joint transition law need not be a product law. Injectivity means only that their simultaneous observations recover the vertex exactly.

An individual coordinate is a classical binary strong lumping; see Geiger and Temmel [2]. On a regular graph it is equivalently a perfect two-coloring, or equitable two-partition, as studied in [1]. The spectral and product constructions belong to the broader theory of perfect structures [3]. We use these established objects to ask a simultaneous separation question. The terminology and invariant are proposed here as an organizing viewpoint; priority for the invariant or the exact consequences below has not been established by a comprehensive literature review. The mathematical claims are supported by the proofs, independently of any claim of historical novelty.

2 Spectral separation and elementary operations

Write $\pi(f) = \sum_v \pi(v)f(v)$, and define

$$\mathcal{A}(G) = \text{span}_{\mathbb{R}}\{f - \pi(f)\mathbf{1} : f \text{ is autonomous}\}.$$

The zero space is allowed. A space of functions *separates vertices* if, for every $u \neq v$, at least one function in the space has different values at u and v .

Theorem 2.1 (Spectral separation). *An autonomous binary f with parameters (a, b) satisfies*

$$P\left(f - \frac{a}{a+b}\mathbf{1}\right) = (1-a-b)\left(f - \frac{a}{a+b}\mathbf{1}\right), \quad \pi(f) = \frac{a}{a+b}. \quad (2.1)$$

Conversely, every real eigenvector of P with exactly two values is an affine rescaling of an autonomous binary function. In particular, $\text{adim}(G) < \infty$ if and only if the two-valued real eigenvectors separate all vertices. Whenever the dimension is finite,

$$\lceil \log_2 n \rceil \leq \text{adim}(G) \leq \dim \mathcal{A}(G) \leq n-1. \quad (2.2)$$

Proof. The defining condition is exactly $Pf = a\mathbf{1} + (1-a-b)f$. Applying the stationary functional π gives $(a+b)\pi(f) = a$, proving (2.1).

Conversely, normalize the two values of an eigenvector to 0 and 1, obtaining a nonconstant binary f . The eigenvector equation says that Pf is an affine function of f , hence constant on each of its two classes. Its value on class 0 is the cross-class probability a ; its value on class 1 is $1-b$. Thus f is autonomous.

The family of generators of $\mathcal{A}(G)$ separates vertices if and only if its span does. If it separates, choose a basis from these generators. Equality of all basis values at two vertices would force equality

of every value in the span. The corresponding binary coordinates therefore separate, proving the middle bound. Every generator has stationary mean zero, so the space has dimension at most $n - 1$. Finally, d binary coordinates have at most 2^d different codes. $\square$

Corollary 2.2 (Integer spectral obstruction). *If G is q -regular, every autonomous binary coordinate gives an integer adjacency eigenvalue other than q . Consequently, if the adjacency matrix has no integer eigenvalue other than q , then $\text{adim}(G) = \infty$.*

Proof. The numbers of opposite-color neighbors are integers $c_0 = qa$ and $c_1 = qb$. The centered coordinate has adjacency eigenvalue $q - c_0 - c_1$, which is an integer and is smaller than q since $a, b > 0$. $\square$

The obstruction immediately gives $\text{adim}(C_5) = \infty$: the nonprincipal adjacency eigenvalues are $(\sqrt{5} - 1)/2$ and $-(\sqrt{5} + 1)/2$, each twice. At the other extreme, every nonconstant binary function on K_n is autonomous, so

$$\text{adim}(K_n) = \lceil \log_2 n \rceil, \quad \text{adim}(Q_d) = d \quad (d \geq 1). \quad (2.3)$$

For the cube, its usual coordinate bits are autonomous with $a = b = 1/d$; the information bound proves optimality.

Proposition 2.3 (Complement and product laws). *If both G and $\overline{G}$ are connected regular graphs, then $\text{adim}(G) = \text{adim}(\overline{G})$. If G, H are connected regular graphs and both autonomy dimensions are finite, then*

$$\text{adim}(G \square H) \leq \text{adim}(G) + \text{adim}(H).$$

Proof. For a fixed binary partition of a regular graph, autonomy is equivalent to a constant number of opposite-color neighbors within each class. In the complement that number becomes the size of the opposite class minus its original value. Thus the two graphs have exactly the same autonomous binary functions.

If G, H have degrees q, s , respectively, lift a coordinate of G to $(x, y) \mapsto f(x)$. Its two flip probabilities are multiplied by $q/(q + s)$ on the Cartesian product, so it remains autonomous. Coordinates of H lift in the same way with factor $s/(q + s)$. Combining separating families from both factors separates the product. $\square$

There is also a finite exact procedure. Enumerate the nonconstant binary functions up to complementation, retain the autonomous ones, and associate to each the unordered vertex pairs that it separates. If some pair is uncovered, the dimension is infinite. Otherwise the dimension is the minimum number of these sets covering all pairs. This is an exact finite set-cover formulation, not a claim of an efficient algorithm or a complexity hardness result.

3 An asymptotically sharp cost of autonomy

Let W_r be the graph obtained from r copies of K_4 by identifying one vertex of each copy to a common hub o . Removing o leaves r disjoint triangles. There are $3r + 1$ vertices; the hub has degree $3r$ and each outer vertex has degree 3.

For $0 \leq p \leq 1$, put

$$H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p), \quad h = H_2(1/3) = \log_2 3 - \frac{2}{3},$$

with $0 \log_2 0 = 0$. Numerically, $h = 0.918295834 \dots$

Theorem 3.1 (Sharp asymptotic overhead). *For every integer $r \geq 1$, the autonomy dimension of W_r is finite and*

$$\left\lceil \frac{\log_2(3r)}{h} \right\rceil \leq \text{adim}(W_r) \leq \min \left\{ 3k : k \geq 1, \binom{3k}{k} \geq 9r \right\}. \quad (3.1)$$

Consequently, as $r \rightarrow \infty$,

$$\text{adim}(W_r) = \frac{\log_2(3r)}{h} + O(\log \log r), \quad (3.2)$$

$$\frac{\text{adim}(W_r)}{\log_2 |V(W_r)|} \rightarrow \frac{1}{h} = 1.088973686 \dots \quad (3.3)$$

In particular, $\text{adim}(W_r) - \lceil \log_2 |V(W_r)| \rceil$ tends to infinity.

Lemma 3.2 (Coordinates on the windmill). *Orient an autonomous binary coordinate so that $f(o) = 0$. Its class 1 consists of exactly k vertices from each outer triangle, for one common $k \in \{1, 2, 3\}$. Conversely, every such choice is autonomous, with*

$$a = k/3, \quad b = (4 - k)/3.$$

Proof. Some outer vertex is selected because the function is nonconstant. The hub therefore has positive flip probability. If a triangle had no selected vertex, its zero vertices would have flip probability zero, a contradiction. Let $k_i \geq 1$ be the number selected in triangle i . A selected vertex there has $4 - k_i$ zero neighbors, including the hub, out of degree 3. Constancy on selected vertices forces all k_i to agree. Conversely, if they all equal k , the hub has flip probability $rk/(3r) = k/3$, every outer zero has k selected neighbors, and every selected vertex has $4 - k$ zero neighbors. When $k = 3$ the hub is the only zero vertex, so the same formulas still hold. $\square$

Lemma 3.3 (Entropy obstruction). *Let Z have any probability distribution on the vertices of a graph. If every autonomous binary f satisfies $H(f(Z)) \leq \eta$, then every separating family of d autonomous coordinates satisfies $H(Z) \leq d\eta$, where entropies are in bits.*

Proof. Injectivity gives $H(Z) = H(f_1(Z), \dots, f_d(Z))$. The entropy chain rule and the fact that conditioning cannot increase entropy give $H(f_1(Z), \dots, f_d(Z)) \leq \sum_j H(f_j(Z)) \leq d\eta$. No independence assumption is used. $\square$

Proof of Theorem 3.1. Choose Z uniformly among the $3r$ outer vertices. After complementing any coordinate if necessary, Lemma 3.2 gives $\Pr(f(Z) = 1) \in \{1/3, 2/3, 1\}$. Hence $H(f(Z)) \leq h$, and Lemma 3.3 proves the lower bound.

For the upper bound, fix $k \geq 1$. Form a hypergraph whose vertices are the k -element subsets of $[3k]$ and whose hyperedges are the unordered partitions of $[3k]$ into three such subsets. It has

$$B = \binom{3k}{k} \text{ vertices,} \quad \Delta = \frac{1}{2} \binom{2k}{k} \text{ edges through each vertex,}$$

and therefore $B\Delta/3$ hyperedges. Take a maximal matching M of pairwise vertex-disjoint hyperedges. Every hyperedge meets a member of M , while each member meets at most 3Δ hyperedges. Thus

$$\frac{B\Delta}{3} \leq 3\Delta|M|, \quad |M| \geq \frac{B}{9}.$$

If $B \geq 9r$, select r of these partitions. In triangle i , assign the three incidence words of the three blocks of partition i , and assign the all-zero word to the hub. All $3r$ outer words are distinct because the hyperedges are disjoint as sets of blocks; each word is nonzero. Every coordinate selects exactly one vertex per triangle. By Lemma 3.2, every coordinate is autonomous. This gives a separating code of length $3k$ and proves finiteness and the upper bound.

Stirling's formula gives

$$\log_2 \binom{3k}{k} = 3kh - \frac{1}{2} \log_2 k + O(1).$$

For $L = \log_2(3r)$, choosing $k = \lceil (L + \log_2 L + C)/(3h) \rceil$ with a fixed sufficiently large C makes $\binom{3k}{k} \geq 9r$ for all sufficiently large r . The upper bound is therefore $L/h + O(\log L)$, which together with the lower bound proves (3.2). The ratio follows since $\log_2(3r+1) = \log_2(3r) + o(1)$. Finally $1/h > 1$, so even the lower bound exceeds $\lceil \log_2(3r+1) \rceil$ by a quantity tending to infinity. $\square$

The coefficient in (3.3) identifies a quantitative cost: the allowed coordinates convey at most $h < 1$ bit each under a natural vertex distribution. The construction shows that this entropy cost is asymptotically attainable. Thus autonomy dimension contains information that the order of a graph alone cannot express.

Proposition 3.4 (The smallest windmill obstruction). *For two copies of K_4 sharing a vertex, $\text{adim}(W_2) = 4$, although $\lceil \log_2 7 \rceil = 3$.*

Proof. Suppose that three coordinates suffice, and complement them to make the hub 000. The six outer vertices then receive six distinct nonzero three-bit words. There are seven such words, so exactly one nonzero word w is omitted. In each column, Lemma 3.2 requires equally many ones in the two triangles, hence an even total. But all seven nonzero words contain four ones in each column, so removing any nonzero w leaves an odd total in a column where w is one. This is impossible.

For a four-coordinate construction use hub 0000, and assign the words

$$\{1000, 0100, 0011\}, \quad \{0010, 0001, 1100\}$$

to the two triangles. Each column selects exactly one vertex of each triangle, and all seven words are distinct. $\square$

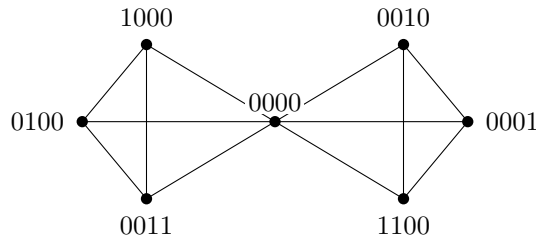


Figure 1: An optimal autonomous code for W_2 . Each column has flip probabilities $a = 1/3$ and $b = 1$. Four coordinates are necessary.

4 A product can create finite autonomy dimension

The product bound of Proposition 2.3 does not have a converse: the product can have finite dimension even if both factors have infinite dimension.

Theorem 4.1 (Product activation). *For the Cartesian product of two five-cycles,*

$$\text{adim}(C_5) = \infty, \quad \text{adim}(C_5 \square C_5) = 6.$$

More precisely, writing its vertices as $(x, y) \in \mathbb{F}_5^2$, every autonomous binary coordinate depends solely on one of the two quantities

$$u = x + 2y \pmod{5}, \quad v = x - 2y \pmod{5}. \quad (4.1)$$

Conversely, every nonconstant binary function of either quantity is autonomous.

Proof. The assertion about C_5 follows from Corollary 2.2. On the Cartesian product, the four steps $(\pm 1, 0), (0, \pm 1)$ change u by $\pm 1, \pm 2$, each with probability $1/4$. They therefore send u to each of its other four values uniformly. The same holds for v . Each projection has the transition law of simple random walk on K_5 . All binary partitions of K_5 are autonomous, proving the converse claim. The transformation (4.1) is invertible over $\mathbb{F}_5$ because its determinant is $-4 \neq 0$. Encoding u with three bits and v with three more therefore gives an injective autonomous six-bit code.

To prove optimality we classify all possible coordinates. Put $\alpha = (\sqrt{5}-1)/2$ and $\beta = -(\sqrt{5}+1)/2$. The adjacency spectrum of C_5 is $2, \alpha, \alpha, \beta, \beta$. The Cartesian adjacency matrix is $A \otimes I + I \otimes A$; tensor products of eigenvectors give all pairwise eigenvalue sums. Thus the product spectrum is

eigenvalue	4	$2 + \alpha$	$2 + \beta$	2α	2β	-1
multiplicity	1	4	4	4	4	8.

The only integer nonprincipal eigenvalue is -1 . By Corollary 2.2, the centered version of every autonomous coordinate lies in its eight-dimensional eigenspace.

Every mean-zero function of u is a -1 adjacency eigenvector: the sum over neighbors is the sum over the four other u -values, which is the negative of the current value. These functions form a four-dimensional space. The same is true for v , and the spaces have zero intersection, because (u, v) ranges over all of $\mathbb{F}_5^2$. Together they exhaust the -1 eigenspace. Hence an autonomous binary f has the form

$$f(x, y) = A(u) + B(v),$$

where a constant can be absorbed into A .

If both A and B were nonconstant, choose values $A_- < A_+$ and $B_- < B_+$. The three distinct sums $A_- + B_- < A_- + B_+ < A_+ + B_+$ all occur, contradicting the fact that f has only two values. Thus f depends on only one of u, v .

For any fixed v , the five possible u -values can only be separated by the coordinates depending on u , of which at least three are required. Likewise at least three coordinates depending on v are necessary. A nonconstant coordinate cannot depend solely on both, so these counts add, giving the lower bound six. $\square$

The graph has only 25 vertices, so its unconstrained binary labeling cost is five. The sixth coordinate is forced by the two separate five-valued observables. At the same time, neither original cycle admits even one nonconstant autonomous coordinate. The phenomenon is therefore not explained by simply lifting coordinates from the factors.

5 Exact classification of cycles

Theorem 5.1. *For $n \geq 3$,*

$$\text{adim}(C_n) = \begin{cases} 2, & n = 3, 4, \\ 3, & n = 6, \\ 4, & n = 12, \\ \infty, & \text{otherwise.} \end{cases}$$

In fact, all autonomous binary coordinates together identify vertices $i, j \in \mathbb{Z}/n\mathbb{Z}$ exactly when $i \equiv j \pmod{\gcd(n, 12)}$.

Proof. For each color b , the constant number c_b of opposite-color neighbors is either one or two. If $(c_0, c_1) = (2, 2)$, the colors alternate with period two. If $(c_0, c_1) = (1, 2)$, every one is isolated and every zero has exactly one zero neighbor, so every zero run has length two; the pattern is a rotation of repeated 001. The reversed pair is its complement. If $(c_0, c_1) = (1, 1)$, every run has length two and the pattern is a rotation of repeated 0011. These exhaust all possibilities, with each period required to divide n .

The period-two patterns distinguish residues modulo two. All rotations of the period-three patterns distinguish residues modulo three, and all rotations of the period-four patterns distinguish residues modulo four. Together, the available patterns distinguish residues modulo the least common multiple of the available periods in $\{2, 3, 4\}$, taking this to be one if none is available. This least common multiple is $\gcd(n, 12)$. Thus separation is possible exactly for $n = 3, 4, 6, 12$.

For these four sizes, the information bound gives the stated lower bounds. For C_3 , use any two different singleton indicators. For C_4 , use the two rotated patterns with codes 00, 01, 11, 10 in cyclic order. For C_6 , use parity and two singleton indicators modulo three. For C_{12} , use two singleton indicators modulo three and two rotated period-four patterns. The residue pairs separate by the Chinese remainder theorem, giving the required numbers of coordinates. $\square$

6 The scope of exact autonomy

The preceding results exploit exact equalities of transition probabilities. The next theorem makes this limitation precise. Give each edge e a positive symmetric weight w_e , and use transition probabilities $P_w(x, y) = w_{xy} / \sum_{z \sim x} w_{xz}$. Definition 1.1 then uses weighted cross-edge fractions.

Theorem 6.1 (Generic weight obstruction). *Fix a connected simple support graph with $n \geq 3$. For Lebesgue-almost every vector of positive edge weights, the weighted walk has infinite autonomy dimension. More precisely, outside a set of measure zero, a nonbipartite support has no nonconstant autonomous binary coordinate, and a bipartite support has only its proper bipartition, up to complement.*

Proof. Fix a nonconstant binary partition that is not a proper coloring. Some monochromatic connected component has at least two vertices. Since the whole graph is connected and both colors occur, that component contains a boundary vertex x with a neighbor z of the opposite color. It also contains a same-color neighbor of x . Choose any other vertex y in the component, so x and y have the same color.

Keep all weights except $w_{xz} = t$ fixed. The cross-color probability at x is $(t + C)/(t + C + D)$ with $D > 0$, hence is strictly increasing in t . The corresponding probability at y does not depend on t . Equality of these two probabilities, which autonomy requires, becomes a nonzero polynomial

equation in the edge weights after clearing the positive denominators. Its solution set has Lebesgue measure zero. There are only finitely many binary partitions, so the union of the exceptional sets for all nonproper partitions still has measure zero.

A proper binary coloring always has both flip probabilities one. A connected graph has such a coloring exactly when it is bipartite, and then the coloring is unique up to complement. A single binary partition cannot separate $n \geq 3$ vertices. The result follows. $\square$

This theorem does not contradict stability under a common rescaling of all edge weights, or under symmetry-preserving changes. It concerns arbitrary variation in the full positive edge-weight space. Exact autonomy dimension measures exact organization of a transition law; an approximate version would require a separate definition and separate theorems.

7 Finite verification and further questions

The companion program `verify_autonomy_dimension.py` uses integer cross-multiplication to enumerate all autonomous cuts up to complement, and an exact minimum set-cover search to calculate finite dimensions. It uses NetworkX for graph generation and rational arithmetic for ranks. The results are stored in `autonomy_verification_results.json`. Running the script with Python and NetworkX reproduces the following checks.

Order	Connected graphs	adim = 1	adim = 2	adim = 3	adim = 4
2	1	1	0	0	0
3	2	0	1	0	0
4	6	0	3	0	0
5	21	0	0	2	0
6	112	0	0	7	0
7	853	0	0	3	2

Table 1: Exact enumeration of the 995 connected unlabeled graphs on two through seven vertices. The 19 graphs counted in the dimension columns have finite dimension; all other 976 have infinite dimension.

The checks also verify the spectral upper bound and regular complement invariance on the applicable graphs in this range; exhaust all binary cuts on C_n for $3 \leq n \leq 18$; and confirm dimensions 2, 4, 4 for W_1, W_2, W_3 . For $C_5 \square C_5$, exact rational rank calculations verify the eight-dimensional -1 eigenspace and the absence of other nonprincipal integer eigenvalues. A minimum set-cover search on the 30 cuts classified in Theorem 4.1 returns six. This last enumeration uses the proved classification; it is not an exhaustive search over all binary functions on 25 vertices.

Finally, the greedy matching construction in the proof of Theorem 3.1, with $k = 1, 2, 3, 4$, produces separating codes of lengths 3, 6, 9, 12 for $W_1, W_5, W_{25}, W_{156}$, respectively. All coordinate transition probabilities and code injectivity are checked exactly. These finite checks supplement the proofs; they do not establish the asymptotic or measure-theoretic statements.

The theory suggests several specific questions. Determine $\text{adim}(W_r)$ exactly, beyond the asymptotic formula. Characterize the pairs of regular graphs with infinite factor dimensions and finite product dimension. Find efficient recognition algorithms for graph classes with few distinct eigenvalues. For an approximate theory, one could bound the variation of cross-color probabilities within each class and ask how the minimum separating family changes with the tolerance. These

questions concern a well-defined invariant with spectral obstructions, explicit constructions, and a sharp asymptotic separation from ordinary binary labeling.

References

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